CS641 Assignment 2

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Solution for Q3

(a) If a cyclic group has n elements then show that it has $\phi(n)$ generators.

Let us consider the cyclic group $G = (S, *)$. Let g be a generator of the group, i.e., $S = \{g, g^2, g^3, ..., g^n\}$, where $g^n = e$ (identity element of the group). Since G is a cyclic group, atleast one such g exists.

Some important points to note -

1. An element h is called generator of $G = (S, *)$ if $\forall x \in S$, $\exists k$, such that $h^k = x$.
2. For any generator g, $g^n = e \implies \forall q \in \mathbb{Z}$ and $r \in \{0, .., n-1\}$, $g^{qn+r} = (g^n)^q * g^r = e^q * g^r = g^r$, i.e., for any k, $g^k = g^{(k \pmod n)}$.
3. Since, all elements of S are unique, and g is a generator, $g^k = e$ is only possible when k is a multiple of n.
4. For any function $f : A \to B$, if $|A| = |B| < \infty$, then f is one-one $\iff$ f is onto.

Consider the function $f : N = \{1, ..., n\} \to S$ given by $f(x) = (g^i)^x$.

Claim : f is one-one $\iff$ i is co-prime to n, i.e., $gcd(i, n) = 1$,where $i \in N$.

Proof:
(Part 1) f is one-one $\implies$ i is co-prime to n.
This is equivalent to prove that i is not co-prime to $n \implies f$ is not one-one. Suppose $gcd(i, n) = d \geq 2$. Let $i = \alpha d$, $n = \beta d$, $gcd(\alpha, \beta) = 1$. Note that if $gcd(\alpha, \beta) > 1$, then we can get a divisor of i, n which is greater than d.
Now, since $2 \leq d \leq i \leq n$, we get $1 \leq \beta = \frac{n}{d} \leq \frac{n}{2} < n$ and $1 \leq \alpha = \frac{i}{d} \leq \frac{n}{d} \leq \frac{n}{2} < n$.
Let us choose $k_1 = \beta$ and $k_2 = n$. Note that $1 \leq k_1, k_2 \leq n$ and $k_1 \neq k_2$, but $f(k_1) = (g^i)^{k_1} = g^{i\beta} = g^{\alpha d \beta} = g^{\alpha(\beta d)} = g^{\alpha n} = e = g^{in} = (g^i)^n = (g^i)^{k_2} = f(k_2)$. Therefore, f is not one-one

(Part 2) i is co-prime to $n \implies f$ is one-one.
We can prove this by contradiction. Assume i and n are co-prime, but f is not one-one, i.e., $\exists k_1, k_2 \in N$, such that, $k_1 \neq k_2$, but $f(k_1) = f(k_2)$. Now, $f(k_1) = f(k_2) \implies (g^i)^{k_1} = (g^i)^{k_2} \implies g^{i(k_1 - k_2)} = e$. Therefore, $i(k_1 - k_2)$ must be a multiple of n (see point 3 above).
Since, $gcd(i, n) = 1$, $(k_1 - k_2)$ must be a multiple of n.
Now, $1 \leq k_1, k_2 \leq n \implies -(n-1) \leq (k_1 - k_2) \leq (n-1) \implies (k_1 - k_2) = 0$, since 0 is the only multiple of n between $-(n-1)$ and $(n-1)$, Therefore, we have $k_1 = k_2$, which is a contradiction to our assumption that $k_1 \neq k_2$.

From (4) and Claim, we can say that $i \in N$ is co-prime to $n \iff f$ is onto. Now, f is onto means that $\forall x \in S, \exists k \in N$ such that $f(k) = (g^i)^k = x \implies g^i$ is a generator of G. Therefore, $i \in N$ is co-prime to $n \iff g^i$ is a generator of G.
Now, number of generators of G = number of $i \in N$ such that g^i is generator of G = number of $i \in N$ co-prime to $n = \phi(n)$. Therefore number of generators of $G = \phi(n)$.
Hence proved.